CNC PROGRAM SHEET

Part: Part | Rev: A | Date: 2026-05-14

Programmer: CNC-AI

Material: ALUMINIUM

Max RPM: 10000 Coolant: through spindle

Setups: 5

Part envelope: 18.4 x 40.1 x 62.3 mm

COORDINATE SYSTEM & WORK ZERO (G54)

Work zero convention	--	**Part height (Y)**	-- mm
CAD up-axis	--	**X range (machine)**	-- to -- mm
Work zero in CAD	X=0 Y=0 Z=0	**Z range (machine)**	-- to -- mm
G54 meaning	Set controller G54 offset so that X0 Y0 Z0 = top face center of part	**Safe rapid Y**	10 mm above work zero

TOOL LIST

T#	Tool ID	Description	Dia (mm)	Manufacturer	Grade	Used for
T01	EM_2FL_3MM	2-flute solid carbide end mill 3mm â€” aluminium geometry	3.0	Sandvik Coromant	1630	contour_mill
T02	SPOT_090_4MM	90-degree solid carbide spot drill, 4mm shank	4.0	Sandvik Coromant	H10F	spot_drill
T03	SLOT_4MM	2-flute solid carbide slot mill 4mm â€” short cut length for stability	4.0	Sandvik Coromant	1630	slot_mill
T04	DRILL_6.0MM	Solid carbide jobber drill 6.0mm	6.0	Sandvik Coromant	H10F	twist_drill
T05	EM_2FL_16MM	2-flute solid carbide end mill 16mm â€” aluminium geometry	16.0	Sandvik Coromant	1630	counterbore_mill
T06	FACEMILL_40MM	40mm indexable face mill â€” 6 inserts	40.0	Sandvik Coromant		face_mill

SETUP 1 -- -Y approach (bottom — part flipped upside down)

PRINCIPAL | 23 clusters | 30 operations

Fixture	Step-jaw vise or fixture plate with bosses as datums.
Description	Flipped setup — part inverted so bottom face is accessible to spindle. Clamp on previously machined top features or use step-jaw vise.
Rotation	None -- standard orientation
WCS	G54
Origin X/Y/Z	X=CENTER (X=-182.941mm, 9.19mm from xmin edge) Y=CENTER (Z=0.000mm, 31.15mm from zmin edge) Z=TOP
Face (W x H)	18.4 x 62.3 mm
Max depth	40.1 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL COUNTERBORE	T02	4.0	0	2.0	10000	600	2.0	--	--	00:01
2	ALU 6 DRILL COUNTERBORE	T04	6.0	0	6.49	9550	716	6.493	--	--	00:01
3	ALU 16 ENDMILL COUNTERBORE RT	T05	16.0	0	6.49	9950	995	6.493	0.1	0.1	00:03
4	ALU 16 ENDMILL COUNTERBORE FINISH	T05	16.0	0	6.49	9950	995	6.493	0.0	0.0	00:03
5	ALU CENTER DRILL	T02	4.0	0	2.0	10000	600	2.0	--	--	00:01
6	ALU 6 DRILL	T04	6.0	0	2.0	9550	716	2.0	--	--	00:01
7	ALU CENTER DRILL COUNTERBORE	T02	4.0	0	2.0	10000	600	2.0	--	--	00:01
8	ALU 6 DRILL COUNTERBORE	T04	6.0	0	6.49	9550	716	6.493	--	--	00:01
9	ALU 16 ENDMILL COUNTERBORE RT	T05	16.0	0	6.49	9950	995	6.493	0.1	0.1	00:03
10	ALU 16 ENDMILL COUNTERBORE FINISH	T05	16.0	0	6.49	9950	995	6.493	0.0	0.0	00:03
11	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
12	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
13	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
14	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
15	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
16	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
17	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
18	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
19	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
20	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
21	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
22	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
23	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
24	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
25	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
26	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
27	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
28	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
29	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
30	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04

ESTIMATED SETUP TIME: 02:11 (includes tool changes)

SETUP 2 -- +Y approach (top — default VMC position)

PRINCIPAL | 3 clusters | 5 operations

Fixture	Standard vise or fixture plate, jaws on side faces.
Description	Standard top-face setup. Clamp part with top face accessible to spindle. Machine all features in this setup before repositioning.
Rotation	None -- standard orientation
WCS	G55
Origin X/Y/Z	X=CENTER (X=-182.941mm, 9.19mm from xmin edge) Y=CENTER (Z=0.000mm, 31.15mm from zmin edge) Z=TOP
Face (W x H)	18.4 x 62.3 mm
Max depth	40.1 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 3 ENDMILL BOSS RF	T01	3.0	0	8.38	10000	300	3.0	0.1	0.1	00:09
2	ALU 3 ENDMILL BOSS FINISH	T01	3.0	0	8.38	10000	300	3.0	0.0	0.0	00:04
3	ALU 3 ENDMILL BOSS RF	T01	3.0	0	8.38	10000	300	3.0	0.1	0.1	00:09
4	ALU 3 ENDMILL BOSS FINISH	T01	3.0	0	8.38	10000	300	3.0	0.0	0.0	00:04
5	ALU 4.0 SLOT MILL SLOT	T03	4.0	0	8.62	10000	200	4.0	--	--	00:10

ESTIMATED SETUP TIME: 00:54 (includes tool changes)

SETUP 3 -- +Z approach (front face)

PRINCIPAL | 1 clusters | 2 operations

Fixture	90° angle plate or tombstone fixture.
Description	Front-face setup. Rotate part 90° so front face is accessible to spindle.
Rotation	None -- standard orientation
WCS	G56
Origin X/Y/Z	X=CENTER (X=-182.941mm) Y=CENTER (Y=-42.586mm) Z=TOP
Face (W x H)	18.4 x 40.1 mm
Max depth	62.3 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T02	4.0	0	2.0	10000	600	2.0	--	--	00:01
2	ALU 6 DRILL	T04	6.0	0	8.0	9550	716	8.0	--	--	00:01

ESTIMATED SETUP TIME: 00:17 (includes tool changes)

SETUP 4 -- +X approach (right side face)

PRINCIPAL | 1 clusters | 2 operations

Fixture	90° angle plate or tombstone fixture.
Description	Right-side face setup. Rotate part 90° so right face is accessible. Clamp on bottom and left faces.
Rotation	None -- standard orientation
WCS	G57
Origin X/Y/Z	X=CENTER (Y=-42.586mm) Y=CENTER (Z=0.000mm) Z=TOP
Face (W x H)	40.1 x 62.3 mm
Max depth	18.4 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T02	4.0	0	2.0	10000	600	2.0	--	--	00:01
2	ALU 6 DRILL	T04	6.0	0	4.0	9550	716	4.0	--	--	00:01

ESTIMATED SETUP TIME: 00:17 (includes tool changes)

SETUP 5 -- -X approach (left side face)

PRINCIPAL | 1 clusters | 1 operations

Fixture	90° angle plate or tombstone fixture.
Description	Left-side face setup. Rotate part 90° so left face is accessible. Clamp on bottom and right faces.
Rotation	None -- standard orientation
WCS	G58
Origin X/Y/Z	X=CENTER (Y=-42.586mm) Y=CENTER (Z=0.000mm) Z=TOP
Face (W x H)	40.1 x 62.3 mm
Max depth	18.4 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 4.0 SLOT MILL SLOT	T03	4.0	0	3.87	10000	200	3.866	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

WARNINGS & FLAGGED ITEMS

Type	Cluster	Operation	Detail
RPM CAPPED	C0	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C0	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C1	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C1	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C2	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C2	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C3	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C3	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C4	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C4	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C5	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C28	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C29	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.